Design Phase – Polyline Editor

Design Approach:

The Polyline Editor is designed using a user-centered and interaction-oriented approach. The design emphasizes direct manipulation, clear system feedback, low cognitive load, and efficient task execution. The interface is structured so that users can easily identify available actions, understand the current system state, and interact with graphical objects on the canvas with minimal confusion.
The final design was refined iteratively during implementation, which aligns with the iterative nature of interaction design discussed in Chapter 5.

Interface Structure:

The interface is divided into four main areas:

Header:

The top header displays the application title “Polyline Editor” and a short subtitle describing it as an interactive editing tool.
This provides immediate system identity and gives the user a clear understanding of the application purpose.

Tools Sidebar:

The left sidebar contains the main controls of the system in a vertically arranged tool panel. The tools include:

- Begin
- Move
- Delete
- Refresh
- Clear Canvas
- Quit

Each button includes:

- an icon
- a text label
- a distinct color

This design improves:

- recognition over recall
- visibility of available actions
- learnability for novice users

The active tool is visually highlighted, helping reduce mode confusion.

Information Panel:

Below the tools panel, an information card displays:

- current mode
- current status message
- total number of polylines

This area supports continuous system feedback and improves the visibility of system status.

Examples of feedback include:

- Begin mode active
- Point added
- Canvas refreshed
- Canvas cleared successfully

Main Workspace:

The right side of the interface serves as the primary interaction area and contains:

Shortcut Strip

A horizontal strip above the canvas displays keyboard shortcuts such as:

- B = Begin
- M = Move
- D = Delete
- R = Refresh
- Q = Quit
- Enter = Finish Polyline

This supports both:

- novice users, through visible command hints
- expert users, through shortcut efficiency

Canvas Panel

The canvas is the main interaction space where users create and edit polylines. It supports:

- point creation
- point selection
- point movement
- deletion of vertices
- real-time redrawing

This area is intentionally large to support precision and direct interaction.

Interaction Design

Begin Tool:
The Begin tool allows the user to create a new polyline by clicking on the canvas. Each click creates a new point, and line segments are drawn incrementally between consecutive points.

This supports:

- direct manipulation
- incremental construction
- immediate feedback

The user presses Enter to complete the current polyline.

Move Tool:
The Move tool allows the user to reposition an existing point by selecting the nearest vertex and dragging it to a new location.

This improves:
- flexibility
- error correction
- user control

Hover highlighting helps users identify selectable points before dragging.

Delete Tool:
The Delete tool removes the nearest point from a polyline. After deletion, the system redraws the polyline structure accordingly.
This supports:

- editing efficiency
- error recovery
- minimal interaction effort

Refresh Action:
The Refresh action redraws all stored polylines on the canvas. Its purpose is to maintain consistency between the internal data model and the visual view.

This supports:

- robustness

- visibility of system state
- recovery from visual inconsistencies

Clear Canvas Action:
The Clear Canvas action removes all polylines from the editor and resets the workspace.

This provides:

- fast workspace reset
- user control over the drawing environment
- support for restarting tasks quickly

Since this is a destructive action, it should be described as a high-impact control in the design.

Quit Tool:
The Quit tool disables interaction with the editor.

This supports:

- user control
- clear task termination

Visual Design Decisions
The interface uses a simple and modern visual structure with:

- colored buttons for action differentiation
- rounded containers for grouping related controls
- a clear separation between controls and workspace
- icons combined with labels for improved recognition

Color is used functionally:

- green for Begin
- blue for Move
- red for Delete
- gray for Refresh
- orange for Clear Canvas
- dark tone for Quit

This improves visual hierarchy and helps users quickly identify tool purpose.

Feedback Design

The system provides both visual and textual feedback.

Visual Feedback

- hovered points are highlighted
- selected points are visually emphasized
- lines update immediately during movement

Textual Feedback

- current mode is always shown
- status messages explain action results
- polyline count updates dynamically

These feedback mechanisms reduce uncertainty and strengthen the user's mental model of the system.

HCI Principles Applied

The current design applies the following HCI principles:

Visibility

All important actions are visible in the sidebar and shortcut strip.

Feedback

The system continuously informs users through status messages and visual updates.

Consistency

Button styling, interaction behavior, and layout remain uniform throughout the interface.

Direct Manipulation

Users interact directly with points and lines on the canvas.

Recognition over Recall

Icons, labels, and visible shortcuts reduce the need to memorize commands.

User Control and Freedom

Users can freely switch tools, edit objects, refresh the view, clear the canvas, or quit.

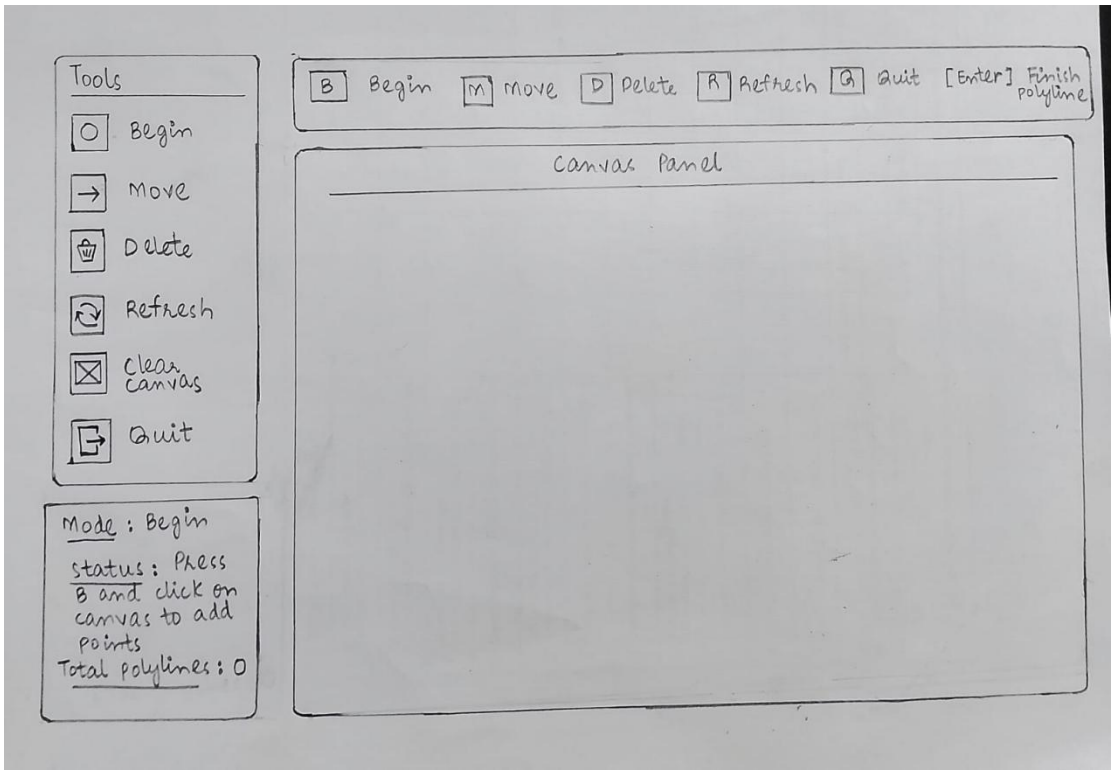
Error Prevention

Hover highlighting and nearest-point detection reduce accidental interactions.

Learnability

The interface is simple, structured, and easy for beginners to understand.

Prototype:



Design Rationale:

The chosen layout is effective because it separates:

- controls on the left
- interaction workspace on the right
- system feedback in a clearly visible info panel
- shortcut guidance above the canvas

This reduces clutter and supports a natural workflow:
select tool → perform action on canvas → observe feedback

The addition of icons, shortcut strip, and clear canvas action makes the design more practical, more interactive, and closer to a real editing system.